

 DATA WAREHOUSE & ANALYTICS

Data Warehouse Analytics Engine

Complete technical walkthrough mapping project architectural notes and SQL scripts to a production-grade Gold Layer data warehouse and business intelligence reporting suite.

 PHASE 01

Data Warehouse Foundation

Establishing the Dimensional Star Schema, automated ingestion pipelines,
and database catalog in Microsoft SQL Server.

Star Schema DDL & Bulk ETL Pipeline

Architectural Design

The Gold schema implements an optimized Dimensional Star Schema designed for high-concurrency analytical queries:

- ▶ **gold.dim_customers:** Master record with demographics, birthdates, and geography.
- ▶ **gold.dim_products:** Product hierarchies, categories, cost basis, and lifecycle dates.
- ▶ **gold.fact_sales:** Central fact tracking sales amount, quantity, order dates, and foreign surrogate keys.

Fast staging with TRUNCATE and native BULK INSERT with TABLOCK optimizations.

```
-- Database & Schema Deployment CREATE DATABASE
DataWarehouseAnalytics; CREATE SCHEMA gold; -- Fact Sales
Definition CREATE TABLE gold.fact_sales( order_number
nvarchar(50), product_key int, customer_key int, order_date
date, sales_amount int, quantity tinyint, price int ); --
High-speed Bulk Ingestion BULK INSERT gold.fact_sales FROM
'/data/gold.fact_sales.csv' WITH (FIRSTROW = 2,
FIELDTERMINATOR = ',', TABLOCK);
```


Customer Behavioral **Segmentation**



LIFESPAN $\geq$ 12M | SPEND $>$ €5K

VIP Tier

High-value long-term accounts with over 12 months engagement and gross revenue exceeding €5,000. Prioritized for loyalty programs.

$\text{Spend} > 5000 \wedge M \geq 12$



LIFESPAN $\geq$ 12M | SPEND $\leq$ €5K

Regular Tier

Established repeat customers retaining continuous tenure over 12 months with disciplined, lower-tier annual cumulative spending.

$\text{Spend} \leq 5000 \wedge M \geq 12$



LIFESPAN $<$ 12 MONTHS

New Customers

Recent acquisitions with first transaction history under 12 months regardless of total basket size. Targeted for onboarding & conversion.

$\text{Lifespan} < 12 \text{ Months}$

Consolidated Customer Report View

▼ Multi-Stage CTE Pipeline

View `gold.report_customers` uses modular CTE decomposition to generate multidimensional analytics:

- ▶ **Stage 1 (base_query):** Flattens sales with customer dimensions and calculates exact customer age via `DATEDIFF(year, birthdate, GETDATE())`.
- ▶ **Stage 2 (customer_aggregation):** Pre-aggregates totals for orders, revenue, product breadth, and tenure lifespan in months.
- ▶ **Stage 3 (KPI Evaluation):** Categorizes demographics into 5 age bands and computes customer lifetime metrics.

```
CREATE VIEW gold.report_customers AS WITH base_query
AS ( SELECT f.order_number, f.sales_amount, c.customer_key,
CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
DATEDIFF(year, c.birthdate, GETDATE()) age FROM
gold.fact_sales f LEFT JOIN gold.dim_customers c ON
c.customer_key = f.customer_key WHERE order_date IS NOT
NULL ), customer_aggregation AS ( SELECT customer_key,
COUNT(DISTINCT order_number) total_orders,
SUM(sales_amount) total_sales, DATEDIFF(month,
MIN(order_date), MAX(order_date)) lifespan FROM base_query
GROUP BY customer_key ) ...
```


Derived Customer KPI Formulas

AOV

Average Order Value

Calculates revenue efficiency per checkout transaction with zero-division safeguard.

$$AOV = \frac{Sales_{total}}{Orders_{total}}$$

AMS

Average Monthly Spend

Quantifies spending velocity normalized over the customer's total active tenure.

$$AMS = \frac{Sales_{total}}{Lifespan \text{ (months)}}$$

REC

Recency In Months

Identifies churn hazard by evaluating months elapsed since the most recent order date.

$$Recency = \Delta t \text{ (Date_{last} , Now)}$$

Product Catalog Cost Segmentation

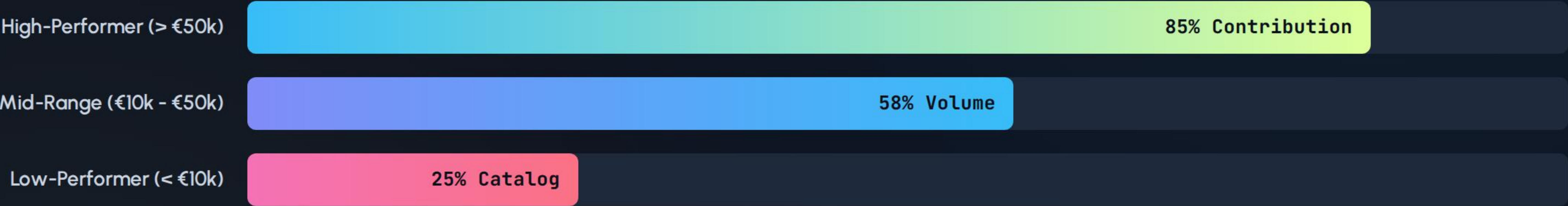
Catalog Tiering Strategy

Product cost distribution is partitioned into operational tiers to structure pricing strategy and margin control:

- ▶ **Below 100:** Budget accessories and high-turnover consumables.
- ▶ **100 - 500:** Mid-tier core merchandise driving steady volume.
- ▶ **500 - 1000:** Premium components and specialized hardware.
- ▶ **Above 1000:** Enterprise-grade flagships and high-margin assets.

```
WITH Product_Segmentation AS ( SELECT product_key,  
product_name, cost, CASE WHEN cost < 100 THEN 'Below 100'  
WHEN cost BETWEEN 100 AND 500 THEN '100-500' WHEN cost  
BETWEEN 500 AND 1000 THEN '500-1000' ELSE 'Above 1000' END  
Cost_Range FROM gold.dim_products ) SELECT Cost_Range,  
COUNT(product_key) AS Total_Products FROM  
Product_Segmentation GROUP BY Cost_Range ORDER BY  
Total_Products DESC;
```


Product Performance Revenue Tiers



Classified within gold.report_products. High-performing SKUs generate over 85% of total gross margin, signaling prime candidates for inventory prioritization.

Cumulative Growth & Moving Averages



Window function query executing `SUM(Total_Sales) OVER (ORDER BY Order_Date)` demonstrates sustained exponential cumulative revenue trajectory.

Year-over-Year Performance Analysis

📈 Window Analytic Logic

Advanced statistical comparison using SQL window functions across partitioned product streams:

- ▶ **Product Benchmark:** `AVG(Current_Sales) OVER(PARTITION BY product_name)` computes product lifetime average sales.
- ▶ **Variance Indicator:** Determines if annual performance is Above Avg or Below Avg.
- ▶ **YoY Delta:** Evaluates year-over-year momentum using `LAG(Current_Sales)` to tag Increase or Decrease.

```
SELECT Order_Year, product_name, Current_Sales,
AVG(Current_Sales) OVER(PARTITION BY product_name) AS
Avg_Sales, Current_Sales - AVG(Current_Sales)
OVER(PARTITION BY product_name) Diff_Avg, CASE WHEN
Current_Sales > AVG(Current_Sales) OVER(PARTITION BY
product_name) THEN 'Above Avg' ELSE 'Below Avg' END AS
Change_Avg, LAG(Current_Sales) OVER(PARTITION BY
product_name ORDER BY order_year) Py_Sales, CASE WHEN
Current_Sales > LAG(Current_Sales) OVER(...) THEN
'Increase' ELSE 'Decrease' END AS Py_Avg FROM
Yearly_Product_Sales;
```


Reporting Suite Catalog Architecture

Object Name	Type	Granularity	Primary KPI Outputs
gold.report_customers	View	Customer Entity	Age Groups, VIP/Regular Segments, AOV, Recency, AMS
gold.report_products	View	Product SKU	Revenue Tiers, Avg Selling Price, AOR, Monthly Run-rate
Time-Series Pipeline	Query	Year / Month	Order Volume, Annualized Quantities, Customer Reach
YoY Product Performance	CTE / Window	Product x Year	YoY Growth %, Prior-Year Delta, Historical Benchmark

✓ ARCHITECTURE COMPLETE

Questions & Discussion

Ready to review query execution plans, indexing strategies, or Power BI semantic model integrations.

 DataWarehouseAnalytics  Gold Schema Engine